

that run critical infrastructure, such as power plants, and transmission and distribution companies, among others. “For now, private sector banks, healthcare and consumer product industries are the most commonly targeted since they are the early adopters of cloud,” says Chintan Matalia, Partner at Deloitte India.

“Earlier, these sectors were unwilling to utilise clouds for their essential data assets due to control issues. But now, even these industries are moving to the cloud,” says Anshuman Sharma, Associate Director CSIRT & Investigative Response, APJ at Verizon Business. By targeting financial services firms, cyber criminals aim to steal sensitive customer information, financial records and transaction data to commit fraud, initiate unauthorised transactions, or demand ransom to restore the data and services they have impacted.

On the other hand, healthcare data can be used to gain access to personal health information, which can then be used for identity theft, insurance fraud, or

after the weakest link in the chain, normally a human, by launching multiple social engineering attacks—which manipulate an individual into divulging sensitive information, such as passwords, access codes, etc. Phishing is another tactic where bad actors hoodwink users into clicking on malicious links or attachments, which can then be further exploited to infect a system. “The attacker will employ tactics such as phishing, or try to get credentials through brute force attacks to unlawfully access user accounts, so they can access [an employee’s] mobile or hack it, and then impersonate [that individual] to create issues,” says Huzefa Motiwala, Director for Systems Engineering, India and SAARC at Palo Alto Networks. Brute force attack uses trial and error to crack passwords, login credentials and encryption keys.

Another common method used by cyber criminals is through misconfiguration. This refers to improper or inadequate set-up of cloud services, resources and se-

THE PROTECTION GAME

Global cloud security software market

2022 **29.3**

Market size (\$ billion)

2028 **39.3**

Indian cloud security software market

2023 **25.39**

Market size (\$ million)

2028 **136.2**

ALL FIGURES ARE ESTIMATES; SOURCE IMARC GROUP

even be sold on the dark web. E-commerce platforms and retailers are also often targeted to access customer payment information, including credit and debit card details, etc. Often, this data is sold on the dark web, from where people can buy it to perpetrate more financial crimes. Such breaches can lead to financial losses for the business as well as its customers, reputational damage, and loss of trust among customers.

Cyber criminals use various methods to gain unauthorised access to cloud accounts. “Most attackers perform extensive reconnaissance, searching for leaked credentials, default or weak passwords, misconfigurations, and human errors,” says Anand Trivedi, Head of APAC at cyber security services provider CyberProof, a UST company. Cyber attackers usually attempt to go

curity settings, which occurs when cloud environments are not configured according to security guidelines. Even poorly secured interfaces and APIs provided by cloud service providers can be exploited by cyber criminals to gain unauthorised access and manipulate cloud resources. Denial of service attacks (DoS) is another method where malicious attempts are made to disrupt the normal functioning of a computer system, network, or online service by overwhelming it with a large volume of traffic or requests, rendering it inaccessible to users.

Both government and private entities need to be wary of these techniques used by attackers. “Additionally, insider threats and third-party vendors are a critical part of the network today, and must be considered when planning defences,” adds Trivedi of CyberProof.